Multisource Compensation Network for Remote Sensing Cross-Domain Scene Classification

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***Abstract*— Cross-domain scene classification refers to the scene classification task in which the training set (termed source domain) and the test set (termed target domain) come from different distributions. Various domain adaptation methods have been developed to reduce the distribution discrepancy between different domains. However, current domain adaptation methods assume that the source domain and target domain share the same categories. In reality, it is hard to find a source domain that can completely cover all the categories of target domain. In this article, we propose to use multiple complementary source domains to form the categories of target domain. A multisource compensation network (MSCN) is proposed to tackle these challenges: distribution discrepancy and category incompleteness. First, a pretrained convolutional neural network (CNN) is exploited to learn the feature representation for each domain. Second, a cross-domain alignment module is developed to reduce the domain shift between source and target domains. Domain shift is reduced by mapping the two domain features into a common feature space. Finally, a classifier complement module is proposed to align categories in multiple sources and learn a target classifier. Two cross-domain classification data sets are constructed using four heterogeneous remote sensing scene classification data sets. Extensive experiments are conducted on these datasets to validate the effectiveness of the proposed method. The proposed method can achieve 81.23% and 81.97% average accuracies on two-source-complementary data set and three-source-complementary data set, respectively.**

***Index Terms*— Cross-domain scene classification, domain adaptation, multisource compensation, remote sensing scene classification.**

I. Introduction

REMOTE sensing scene classification has attracted much attention in many remote sensing applications, such as military [1]–[3] and agriculture [4], [5]. Remote sensing scene classification aims to identify a sematic label to each image with a supervised learning paradigm [6], [7]. A classifier can be trained on a labeled image set (termed training set) and tested on an unlabeled image set (termed test set) [8], [9]. It is assumed that the training set and test set are drawn from the same distribution [10], [11]. However, this assumption is frequently challenged in remote sensing applications [12]. Especially, the remote sensing images may be captured with different sensors and over diverse locations of the ground surface [13]. Changes in sensor types, shooting angles, and illumination condition can cause large distribution differences across images. The task that tries to achieve high classification performance when the training and testing data come from different distributions is known as cross-domain scene classification [14], [15]. A rich line of domain adaptation approaches [13], [16], [17] provide an option, so that the classifier trained on a source distribution (termed source domain) can generalize well on a different target distribution (termed target domain) [18]–[20].

Domain adaptation methods aim to reduce the distribution divergence between source and target domains [21], [22]. Previous domain adaptation methods map source and target domains into a domain-invariant feature space [23]. In this feature space, the source and target domains share the same (or similar) distributions. Recently, many domain adaptation methods extend a deep neural network to learn domain-free features. An adaptation module is embedded into the deep architecture [24]–[26]. Adversarial domain adaptation is a widely used method among deep domain adaptation methods [25]–[27]. Adversarial domain adaptation learns the domain-free features in a two-player game inspired by generative adversarial networks (GANs) [28]. In domain adversarial network, a discriminator is used to distinguish the source features from target features, while a feature extractor learns the domain-free features to confuse the discriminator.

Despite efficacy, existing domain adaptation methods rely on an assumption that source domain and target domain share the same categories, $C_t = C_s$ (C_t denotes target domain categories and C_s denotes source domain categories) [14]–[16]. However, such an assumption is often difficult to satisfy in the cross-domain scene classification. In most cases, the source domain category is a subset of target domain category, i.e., $C_t \supset C_s$. A more realistic scenario is multiple-source domains that are available, and the union of multiple-source domains shares the same categories with target domain, i.e., $C_t = \cup_{k=1}^{K} C_{s_k}$ (C_{s_k} denotes the categories in kth source domain). However, two problems need to be considered in

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using multiple-source domains: 1) how to reduce the domain shift across multiple sources and target domains and 2) how to align the unshared categories among multiple sources. A simple method is to apply the single-source domain adaptation method by combining all source domains into a single-source domain. However, this method learns a mixture distribution for all source domains and cannot capture the distribution for each source domain [29], [30].

To address the problems, in this article, a multisource compensation network (MSCN) is proposed to learn knowledge from multiple complementary source domains. The proposed MSCN composes of three parts: feature representation, cross-domain alignment, and classifier complement. First, a feature extractor is exploited to represent the images in different domains (multiple sources and target domains). Second, a cross-domain alignment module is proposed to reduce the domain shift by mapping different domains into a common feature space. Finally, a classifier complement module is proposed to align categories in multiple sources and learn a target classifier.

In the first stage, a pretrained convolutional neural network (CNN) is used as the feature extractor. Since AlexNet [31] is lightweight and has proven to capture effective information for remote sensing images, it is used to extract features. In the second stage, multiple source-specific domain discriminators are used to compose cross-domain alignment module. Each domain discriminator is responsible for one source domain, which considers the large discrepancy within multiple sources [30]. The distribution discrepancy between source and target domains is reduced by learning a domain-invariant feature that is indistinguishable for a domain discriminator. In the final stage, multiple source-specific classifiers are trained with the corresponding domain-free source features and then combined with different weights to form the target classifier. Aligned target category predictions are achieved by accumulating the results regarding same category. Different weights are considered because distribution similarities between each source and target domains are different.

To sum up, the main contributions of this article are in three aspects.

1) The assumption that the source domain and target domain share the same categories is relaxed. The union of multiple complementary source domains is used to cover the categories of target domain.
2) An effective MSCN is proposed to learn knowledge from multiple complementary source domains. To the best of our knowledge, this is the first article using multiple-source domain adaptation with unshared categories in remote sensing.
3) Cross-domain data sets are constructed for cross-domain scene classification. The cross-domain data sets are composed of four heterogeneous remote sensing classification data sets and can be used to form a multisource compensation scenario.

The rest of this article is organized as follows. In Section II, the related works on remote sensing scene classification are briefly reviewed. In Section III, the proposed method is discussed in detail. In Section IV, experiments are provided. Section V concludes this article.

II. Related Work

This section briefly reviews the related works for remote sensing scene classification and cross-domain scene classification.

A. Remote Sensing Scene Classification

The goal of remote sensing scene classification is to automatically assign a sematic label to the corresponding remote sensing image. A classifier can be trained on a labeled image set and tested on an unlabeled image set. It is assumed that the training set and test set are drawn from the same distribution. Remote sensing scene classification mainly focuses on learning a good scene representation and classifier. Since remote sensing images usually have large sizes and complicated contains, a well-designed method is needed to represent the images [1], [4], [11]. Deep neural networks are good at learning an effective scene representation for remote sensing image [7], [8], [10]. Penatti *et al.* [32] use a pretrained CNN to extract hierarchical features and classify the remote sensing scene by a support vector machine (SVM). Castelluccio *et al.* [33] directly fine-tune the pretrained neural network with labeled images. Recently, many new works are developed to learn more precise and discriminative representations for remote sensing images [4], [6], [10]. For example, Cheng *et al.* [10] regularize the CNN features through metric learning to keep the intraclass similarity. Lu *et al.* [11] use a weighted deconvolution network and spatial pyramid model to keep the structure information of remote sensing images.

Despite efficacy, remote sensing scene classification assumes the training set and test set following the same distribution. When the distribution of training set and test set is different, effective methods need to be proposed to ensure high classification accuracy on the test set.

B. Cross-Domain Scene Classification

A lot of cross-domain scene classification methods have been proposed to generalize well on another differently distributed data [16], [21], [25], [34], [35]. Apart from the feature representation and classifier, the cross-domain scene classification method adds a domain adaptation module to reduce the distribution discrepancy [25], [36]. According to the way the distribution similarity measured, domain adaptation methods can be categorized into two groups: distance metric method and adversarial learning method. The distance metric method attempts to reduce the distribution distance between source and target domains. It maps the source and target domains into a common feature space. In this feature space, different distribution measures are used to regularize the distance of source and target domain features, such as the Wasserstein distance (WD) [37] and maximum mean discrepancy (MMD) [23], [38]. For instance, Long *et al.* [39] map the features to a reproducing kernel Hilbert space and use multikernel MMD to measure

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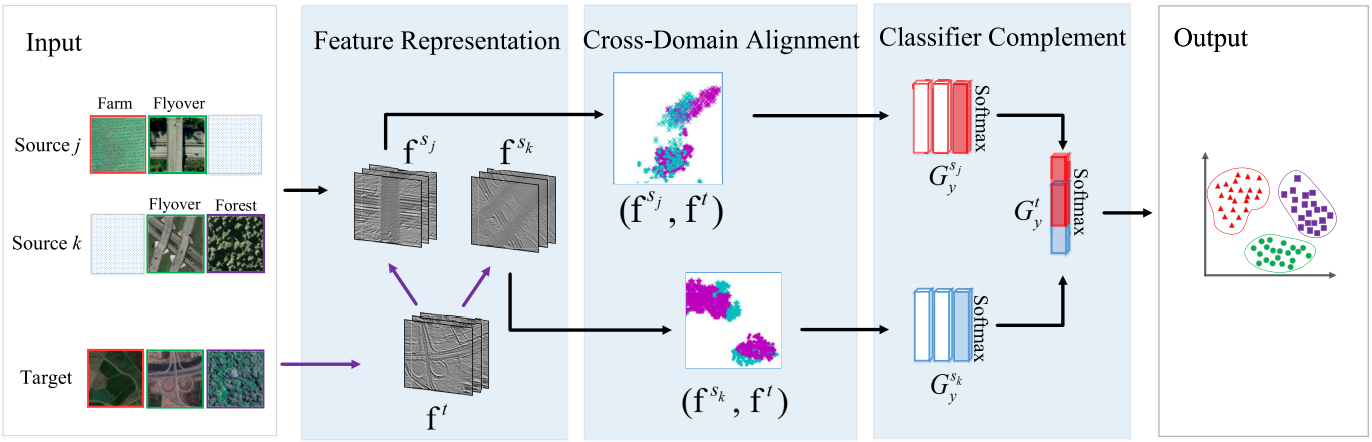


Fig. 1. Framework of the proposed method. Cross-domain scene classification tries to identify the unlabeled target domain with multiple labeled source domains. First, a pretrained CNN is used to extract the image features in different domains. Then, each source domain features and target domain features are aligned with the cross-domain alignment module. Finally, a classifier complement module learns the target classifier by combining multiple source-specific classifiers.

the domain similarity. Adversarial learning learns domain-invariant features through the adversary between domain classifier and feature extractor, such that the domain classifier cannot distinguish which domain the features come from [25], [40]. For example, reverse gradient (RevGrad) [36] learns domain-free features by making the source and target domains indistinguishable for a domain classifier. Domain adversarial neural network (DANN) [27] reduces the domain shift through the adversary of domain discriminator and feature extractor similar to GAN [28]. Siamese-GAN [14] uses an encoder–decoder and discriminator to learn the domain-invariant feature representation. In this article, the widely used adversarial learning is exploited to reduce the data shift between different domains.

All these abovementioned methods only consider the single-source domain. When multiple-source domains are available, the complementary information in different source domains can boost the target domain learning [2], [29]. To use multiple-source domains, adaptive SVM (A-SVM) [41] ensembles multiple source-specific classifiers to form a final target categorization model. Multisource data bring more information for the transfer, but it also brings more challenges. The large distribution divergence across multisource domains need to be considered [42]. Multibranch neural network (MB-Net) [17] proposes to reduce the distribution distance between multiple source–target pairs at feature representation and decision levels. However, MB-Net is only suitable for the case that multiple-source domains, and target domain have the same classes and cannot cope with the situation that target domain has more categories than any single-source domain. The effectiveness of multisource domain adaptation has not been fully explored. In this article, using multiple complementary source domains to finish the target domain classification task is studied.

III. Method

For cross-domain scene classification task with multiple complementary sources, K labeled data sets (source domains) $[S_1, \cdots, S_K]$ and one unlabeled data set (target domain) T are provided. The kth source domain is characterized by a sample set $S_k = \{(x_i^{s_k}, y_i^{s_k})\}_{i=1}^{n_{s_k}}$, where $x_i^{s_k}$ is the ith image, $y_i^{s_k}$ is the corresponding label, and n_{s_k} is the sample size of source domain S_k. Target domain is characterized by data set $T = \{x_i^t\}_{i=1}^{n_t}$, where n_t is the number of unlabeled target domain samples. Every source domain S_k is sampled from distribution P_{s_k} and is associated with categories $\mathcal{C}_{s_k}$. Target domain T is sampled from distribution P_t and is associated with categories $\mathcal{C}_t$. Every source domain may follow different distribution $P_{s_i} \neq P_{s_j}$ and has different distribution to target domain $P_{s_k} \neq P_t$. Any single-source domain categories $\mathcal{C}_{s_k}$ cannot cover target domain categories $\mathcal{C}_t$. However, the union of all source domain categories is the same with target domain categories, $\mathcal{C}_t = \cup_{k=1}^{K} \mathcal{C}_{s_k}$.

The ultimate goal is to predict correct labels $\{y_i^t\}_{i=1}^{n_t}$ for target domain images $\{x_i^t\}_{i=1}^{n_t}$ with the information provided by multiple-source domains $[S_1, \cdots, S_K]$. Since the distribution and category are different for each domain, two problems need to be solved: domain shift and category mismatch. Coping with these problems, three modules are adopted in this article. First, a feature representation module is employed to extract a good scene representation for each domain. Second, a cross-domain alignment module is applied to reduce the domain shift by mapping different domain features to a common feature space. Finally, a classifier complement module is used to learn a target classifier by aligning categories in multiple sources. A clear flowchart is shown in Fig. 1.

A. Feature Representation

We follow the popular approaches for incorporating pretrained CNN as the backbone to learn the scene representations [7], [33]. Specifically, AlexNet [31] that is lightweight and has proven to be effective in extracting remote sensing image features is exploited. The top five convolution layers and three max-pooling layers of AlexNet are used to compose the feature extractor G_f. Activations of the last convolution layer are used as the feature representation $\mathbf{f}_i$. Considering

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that different domains have large discrepancies in size and sampling resolution, preprocessing is used to normalize the input image to same size. The feature extractor is shared for K source and target domains. The function of generating image features is described as follows:

$$\mathbf{f}_i = G_f(x_i, \theta_f) \tag{1}$$

where θ_f denotes the parameters of feature extractor. $\mathbf{f}_i \in \mathbb{R}^D$ is the learned feature, which is obtained by reshaping the activations of the last convolution layer to a D-dimensional feature vector. x_i denotes an image from source domain images or target domain. To distinguish source domain feature and target domain feature, $\mathbf{f}_i^{s_k}$ denotes feature of image in source domain S_k and $\mathbf{f}_i^t$ denotes feature of image in target domain T.

B. Cross-Domain Alignment

To reduce the distribution discrepancy cross multiple sources and target domains, adversarial learning is exploited inspired by DANN [27]. The domain-free features are learned through the adversary of feature extractor G_f and domain discriminator G_d similar to GAN [28]. The discriminator is trained to distinguish source samples from target samples. Feature extractor is trained to learn domain-invariant features to confuse the domain discriminator.

In this article, the feature extractor G_f is the abovementioned feature representation module, which is shared for all the K source domains and target domain. The discriminator G_d is composed of K source-specific discriminators $\{G_d^k\}_{k=1}^K$. Each discriminator is responsible for one source domain. G_d^k receives the corresponding source domain and target domain features as input and classifies which domain the features come from. In the adversarial learning, the parameters θ_d^k of domain discriminator G_d^k are learned by minimizing the adversarial loss $E_k(\theta_d^k, \theta_f)$, and the parameters θ_f of feature extractor G_f are learned by maximizing $E_k(\theta_d^k, \theta_f)$. The adversarial loss is described as follows:

$$\min_{\theta_d^k}\max_{\theta_f} E_k\left(\theta_d^k, \theta_f\right) = -\frac{1}{n_{s_k}}\sum_{i=1}^{n_{s_k}} \log G_d^k\left(G_f\left(x_i^{s_k}\right)\right) - \frac{1}{n_t}\sum_{j=1}^{n_t}\log\left(1 - G_d^k\left(G_f\left(x_j^t\right)\right)\right). \tag{2}$$

In addition, to measure the similarity between each source and target domain, the target–source perplexity scores $\{W_k(x^t; G_f, G_d^k)\}_{k=1}^K$ supplied by K discriminators $\{G_d^k\}_{k=1}^K$ are used. W_k is described as follows:

$$W_k\left(x^t; G_f, G_d^k\right) = -\frac{1}{n_t}\sum_{j=1}^{n_t}\log\left(1 - G_d^k\left(G_f\left(x_j^t\right)\right)\right). \tag{3}$$

A large target–source perplexity score corresponds to a large distribution similarity. Specifically, for an input target domain image, if it has a similar distribution with source domain image, the discriminator G_d^k may misclassify it as a source domain image. Then, $G_d^k(G_f(x_j^t))$ is close to 1. Perplexity score W_k is of a large value.

C. Classifier Complement

Reweighting different source classifiers to boost the target domain learning is a common solution for multisource domain adaptation [43]. However, most existing methods are only suitable for the case that the source domain and target domain share the same categories. Inspired by [30], a classifier complementary module is used to align the categories in multiple sources and learn a target classifier. This module is conducted in two steps. First, K source-specific classifiers $\{G_y^{s_k}\}_{k=1}^K$ are trained with the K source domains. Second, the multiple source-specific classifiers are aligned and combined with different weights to form a target classifier G_y^t.

For the source classifier $G_y^{s_k}$, it is trained with the domain-invariant source features $\mathbf{f}_i^{s_k}$ and corresponding labels $y_i^{s_k}$. Loss function for learning source classifier $G_y^{s_k}$ is described as follows:

$$\min_{\theta_y^{s_k}}\max_{\theta_f} L_k'\left(\theta_y^{s_k}, \theta_f\right) = \frac{1}{n_{s_k}}\sum_{i=1}^{n_{s_k}} \mathcal{L}\left(G_y^{s_k}\left(G_f\left(x_i^{s_k}\right)\right), y_i^{s_k}\right) \tag{4}$$

where $\theta_y^{s_k}$ are the parameters of source classifier learned by minimizing the loss snd $\mathcal{L}(\cdot)$ is the cross entropy loss.

After obtaining the multiple source-specific classifiers, the source classifiers are combined to form the target classifier. Since the different source and target domains have different distribution similarities, the source classifiers should be assigned corresponding weights when composing the target classifier. The target–source perplexity score W_k is used to denote weight. A high perplexity score means large similarity between the source and target domains. The source classifier $G_y^{s_k}$ trained with similar source domain S_k is allocated with high weight.

For a target domain image, it get K predictions from K source classifiers. The probability of image x_i^t belonging to category c_{s_k} predicted by classifier $G_y^{s_k}$ can be denoted as

$$p^{s_k}\left(y_i = c_{s_k}|\mathbf{f}_i^t\right) = G_y^{s_k}\left(G_f\left(x_i^t\right)\right) \tag{5}$$

where $c_{s_k} \in \mathcal{C}_{s_k}$.

If every source domain has same categories with target domain, the probability of image x_i^t belonging to category c_t is the sum of weighted probabilities provided by K source classifiers. The probability of image x_i^t belonging to category c_t can be calculated as follows:

$$\tilde{p}\left(y_i = c_t|\mathbf{f}_i^t\right) := \tilde{G}_y^t\left(G_f\left(x_i^t\right)\right) = \sum_{k=1}^{K}\frac{W_k\left(x^t; G_f, G_d^k\right)}{\sum_{j=1}^K W_j\left(x^t; G_f, G_d^j\right)} p^{s_k}\left(y_i = c_t|\mathbf{f}_i^t\right) \tag{6}$$

where $\tilde{p}(y_i = c_t|\mathbf{f}_i^t)$ is the category probability predicted by target classifier $\tilde{G}_y^t$ with the setting $\mathcal{C}_{s_k} = \mathcal{C}_t, k = 1, \cdots K$. $p^{s_k}(y_i = c_t|\mathbf{f}_i^t)$ is the probability provided by source classifier $G_y^{s_k}$. $W_k(x^t; G_f, G_d^k)p^{s_k}(y_i = c_t|\mathbf{f}_i^t)$ means weighting the probabilities according to the similarity between source and target domains. $\sum_{k=1}^K$ indicates the sum of weighted probabilities provided by K source classifiers. $1/\sum_{j=1}^K W_j(x^t; G_f, G_d^j)$ means normalizing the weights.

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However, since the target domain has more categories than any single-source domain, simply weighting all the predicted probabilities together will lead to category misalignment and incomplete categories. To tackle the category mismatch problem, a constraint is posed on the combination of source classifiers. Only predicted probability corresponding to the same category can be accumulated. The probability of image x_i^t predicted by target classifier G_y^t is described as follows:

$$p\left(y_i = c_t|\mathbf{f}_i^t\right) := G_y^t\left(G_f\left(x_i^t\right)\right) = \sum_{c_t \in \mathcal{C}_{s_k}} \frac{W_k\left(x^t; G_f, G_d^k\right)}{\sum_{c_t \in \mathcal{C}_{s_j}} W_j\left(x^t; G_f, G_d^j\right)} p^{s_k}\left(y_i = c_t|\mathbf{f}_i^t\right) \quad (7)$$

where $\sum_{c_t \in \mathcal{C}_{s_k}}$ means only source classifiers whose source domains containing category c_t contributing to the target classifier. $\sum_{c_t \in \mathcal{C}_{s_j}} W_j(x^t; G_f, G_d^k)$ is the sum of source classifier weights used. In this way, the unshared categories among multiple-source domains are aligned. Target domain images are categorized to category $\mathcal{C}_t = \cup_{k=1}^K \mathcal{C}_{s_k}$.

D. Overall Objective Function of MSCN

The object function is jointly optimized with respect to feature extractor, discriminators, and source classifiers. Moreover, for a better performance on target domain, we optimize the classification loss on target samples with pseudolabels. In specific, for an input target domain image, it is assigned a label according to the predicted probability. If the probability is larger than a threshold γ, the label will be thought as a high confidence label. The high confidence label is treated as a pseudolabel. Target domain images with pseudolabels are used as training data for each source classifier. The classification loss of source classifier $G_y^{s_k}$ is described as follows:

$$\min_{\theta_y^{s_k}} \max_{\theta_f} L_k\left(\theta_y^{s_k}, \theta_f\right) = \frac{1}{n_{s_k}} \sum_{i=1}^{n_{s_k}} \mathcal{L}\left(G_y^{s_k}\left(G_f\left(x_i^{s_k}\right)\right), y_i^{s_k}\right) + \frac{1}{\hat{n}_{s_k}} \sum_{\hat{y}_j \in \mathcal{C}_{s_k}} \mathcal{L}\left(G_y^{s_k}\left(G_f(\hat{x}_j^t)\right), \hat{y}_j\right) \quad (8)$$

where $\hat{x}_j^t$ and $\hat{y}_j$ represent the image and pseudolabel of target sample with high confidence, respectively. $\sum_{\hat{y}_j \in \mathcal{C}_{s_k}}$ means only the predicted label belonging to the categories of source domain will be used to optimize the corresponding classifier. $\hat{n}_{s_k}$ denotes the number of pseudolabels belonging to the source domain categories $\mathcal{C}_{s_k}$.

The overall objective function of MSCN is composed of adversarial loss and classification loss in K domains

$$\min_{\theta_d^k, \theta_y^{s_k}} \max_{\theta_f} \sum_{k=1}^K \left(E_k\left(\theta_d^k, \theta_f\right) + L_k\left(\theta_y^{s_k}, \theta_f\right)\right) = \sum_{k=1}^K \left(-\frac{1}{n_{s_k}} \sum_{i=1}^{n_{s_k}} \log G_d^k\left(G_f\left(x_i^{s_k}\right)\right) - \frac{1}{n_t} \sum_{j=1}^{n_t} \log\left(1 - G_d^k\left(G_f\left(x_j^t\right)\right)\right) + \frac{1}{n_{s_k}} \sum_{i=1}^{n_{s_k}} \mathcal{L}\left(G_y^{s_k}\left(G_f\left(x_i^{s_k}\right)\right), y_i^{s_k}\right) + \frac{1}{\hat{n}_{s_k}} \sum_{\hat{y}_j \in \mathcal{C}_{s_k}} \mathcal{L}\left(G_y^{s_k}\left(G_f\left(\hat{x}_j^t\right)\right), \hat{y}_j\right) \right) \quad (9)$$

where $\sum_{k=1}^K$ means summarizing the loss in K domains. Loss in each domain contains adversarial loss E_k and classification loss L_k.

Algorithm 1 Framework of MSCN

Input: K source domain data sets $[S_1, \cdots, S_K]$; target domain data set T.
Output: Target class labels $\left\{y_i^t\right\}_{i=1}^{n_t}$.

```
Initialize the parameters θ_f, θ_y^{s_k} of feature extractor G_f and every source classifier G_y^{s_k};
repeat
  Learn domain-free features:
  for epoch = 1 : num_epoch_F do
    Compute every source domain and target domain features: f_i^{s_k} and f_i^t;
    Update feature extractor G_f and every discriminator G_d^k by minimaxing the loss defined by Eq. 2.
  end for
  Learn target domain classifier:
  for epoch = 1 : num_epoch_Y do
    Update every source classifier G_y^{s_k} and feature extractor G_f with (f_i^{s_k}, y_i^{s_k}), (f̂_j^t, ŷ_j) by Eq. 8;
    Classify the target domain images by Eq. 7;
    Update the high confidence target domain images (x̂_j^t, ŷ_j).
  end for
until A fixed number of iterations or classification loss of Eq. 4 converges.
return {y_i^t}_{i=1}^{n_t}.
```

The alternative training procedure of MSCN is summarized in Algorithm 1. Training time computational complexity of the proposed method is $O(t_1 \cdot \sum_{l=1}^L (M_l^2 \cdot K_l^2 \cdot C_{l-1} \cdot C_l)) + O(t_2 \cdot \sum_{y=1}^Y (H_{y-1} \cdot H_y)) + O(t_3 \cdot \sum_{d=1}^D (F_{d-1} \cdot F_d))$, where $O(t_2 \cdot \sum_{y=1}^Y (H_{y-1} \cdot H_y))$, $O(t_2 \cdot \sum_{y=1}^Y (H_{y-1} \cdot H_y))$, and $O(t_3 \cdot \sum_{d=1}^D (F_{d-1} \cdot F_d))$ are the complexity of feature extractor, source-specific classifier, and source-specific discriminator, respectively. t_1, t_2, and t_3 are the number of gradient steps in optimizing corresponding module. L is the number of convolution layers. M_l^2, K_l^2, C_{l-1}, and C_l denote feature map size, kernel size, in channels, and out channels, respectively. Y and D are the fully connected layer number of classifier and discriminator, respectively. H_y and F_d are the number of nodes in corresponding yth and dth fully connected layers. Test time computational complexity of the proposed method is $O(\sum_{l=1}^L (M_l^2 \cdot K_l^2 \cdot C_{l-1} \cdot C_l)) + O(\sum_{y=1}^Y (H_{y-1} \cdot H_y))$. Computational complexity depends linearly on the input image such that the proposed method runs within an acceptable computational time.

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IV. EXPERIMENTS

In this section, comprehensive experiments are conducted to assess the suitability of the proposed method. First, the remote sensing scene classification data sets are described in detail. Second, the effectiveness of each module in the proposed method is validated. Finally, the proposed method is compared with some state-of-the-art methods.

A. Data Sets

Since the cross-domain scene contains different domains, four heterogeneous remote sensing classification data sets are used to compose the cross-domain classification data sets. These images are captured by optical sensors and composed of the R-, G-, and B-bands. In addition, our method can also classify the scene of a multispectral image. To tackle the multispectral image, three spectral bands closest to RGB bands are selected to form a pseudocolor image.

1) Remote Sensing Classification Data Set: In this section, four heterogeneous remote sensing classification data sets, Merced [44], AID [45], Northwestern Polytechnical University (NWPU) [46], and PatternNet [47], are selected to build cross-domain scene data sets. These data sets are acquired over different regions with different sensors, thus having large distribution divergence across them.

Merced[1]*:* This data set consists of 21 classes of aerial scenes, with each class composed of 100 images. The image has a size of 256 × 256 pixels and a resolution of 0.3 m. These images are collected from the United States Geological Survey (USGS) National Map over multiple U.S. regions. All the images are derived from optical imagery and composed of the R-, G-, and B-bands.

AID[2]*:* This data set includes 10 000 RGB images with 30 aerial scene categories. These images have a large scale of 600 × 600 pixels with resolution range from 0.5 to 8 m, which are collected from Google Earth imagery. Google Earth imagery is collected by many satellites and aircraft, such as Landsat 8, GeoEye-1, and WorldView-2. The sensor bands of Landsat 8 are blue: 450–515 nm, green: 525–600 nm, and red: 630–680 nm. Sensor bands of GeoEye-1 are blue: 450–510 nm, green: 510–580 nm, and red: 655–690 nm. Sensor bands of WorldView-2 are blue: 450–510 nm, green: 510–580 nm, and red: 630–690 nm.

NWPU[3]*:* This data set is created by NWPU. This data set is composed of 45 scene classes and includes 700 RGB images in each class. Every image has a size of 256 × 256 pixels. These images are extracted from Google Earth imagery.

PatternNet[4]*:* This data set covers 38 scene classes and each class contains 800 RGB images. The size of the image is 256 × 256 pixels. All these images are collected from Google Earth imagery with multiresolutions.

[1]http://vision.ucmerced.edu/data sets/landuse.html
[2]http://captain.whu.edu.cn/project/AID
[3]http://www.escience.cn/people/JunweiHan/NWPU-RESISC45.html
[4]https://sites.google.com/view/zhouwx/data set

TABLE I
CATEGORIES OF MULTIDUPLICATION DATA SET

Class	Merced	AID	NWPU	PatternNet
Airfield	100	360	1400	800
Anchorage	100	380	700	800
Beach	100	400	700	800
Dense Residential	100	410	700	800
Farm	100	370	1400	800
Flyover	100	420	700	800
Forest	100	250	700	800
Game Space	100	660	1400	1600
Parking Space	100	390	700	800
River	100	410	700	800
Sparse Residential	100	300	700	800
Storage Cisterns	100	360	700	800
Total	1200	4710	10500	10400

2) Cross-Domain Classification Data Set: To verify the benefits of the proposed method, the four remote sensing scene classification data sets are selected to compose two-source case and three source case cross-domain data sets. Furthermore, cross-domain data sets are divided into multiduplication data set and multicomplementary data set.

a) Two-source case cross-domain data set: It is composed of two source domains and one target domain. Three data sets, Merced, AID, and NWPU, are used in the two-source case cross-domain data set.

b) Three-source case cross-domain data set: It is composed of three source domains and one target domain. All four data sets are used in three-source case cross-domain data set.

c) Multiduplication data set: It specifies the cross-domain data set with each domain sharing the same categories. Following the setting in [17], 12 common classes among the four mentioned data sets are selected to form the multiduplication data set.

Since different data sets are collected and labeled by different experts, the same scene may be denoted by different names. In this article, new uniform names are used to represent the categories of each data set. Particularly, Playground and Stadium in AID are combined to form Game Space. Airplane and Airport in NWPU are combined to form Airfield. Circular Farmland and Rectangular Farmland in NWPU are combined to form Farm. Ground Track Field and Stadium in NWPU are combined to form GameSpace. Football Field and Tennis Court in PatternNet are combined to form GameSpace. For other categories, there is only a change in name but not in contained images. Some sample images for the four datasets are shown in Fig. 2. A detailed description of multiduplication data set is shown in Table I. The numbers in Table I indicate the amount of images for each class in the corresponding data set. For the two-source case cross-domain data set, each class contains all images the original data sets have. For the three-source case cross-domain data set, each class only randomly selects 100 images.

d) Multicomplementary data set: It is a cross-domain data set with the setting that each source domain only contains partial categories of the target domain, and the union of all source domains has the same categories with target domain.

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TABLE II
TWO-SOURCE-COMPLEMENTARY DATA SET $\{\bar{N}, \bar{A}\} \rightarrow M$

Class	NWPU	AID	Merced
Airfield	1400	- -	100
Anchorage	700	- -	100
Beach	700	400	100
Dense Residential	700	410	100
Farm	1400	370	100
Flyover	700	420	100
Forest	700	250	100
Game Space	1400	660	100
Parking Space	700	390	100
River	700	410	100
Sparse Residential	- -	300	100
Storage Cisterns	- -	360	100
Total	9100	3970	1200

TABLE III
THREE-SOURCE-COMPLEMENTARY DATA SET $\{\bar{N}, \bar{A}, \bar{M}\} \rightarrow P$

Class	NWPU	AID	Merced	PatternNet
Airfield	100	100	100	100
Anchorage	100	100	100	100
Beach	100	100	100	100
Dense Residential	100	100	100	100
Farm	100	100	100	100
Flyover	100	100	100	100
Forest	100	- -	- -	100
Game Space	100	- -	- -	100
Parking Space	- -	100	- -	100
River	- -	100	- -	100
Sparse Residential	- -	- -	100	100
Storage Cisterns	- -	- -	100	100
Total	800	800	800	1200

This data set is used to evaluate the effectiveness of MSCN method on multisource compensation scenario. Multicomplementary data set is constructed based on multiduplication data set.

For the two-source-complementary data set, two domains are randomly selected as the source domains and the remaining one as the target domain. In source domains, the 12 categories are divided into two parts: the shared classes and private classes of each source. Under the category order of multiduplication data set, the first and last n categories are selected as private classes of each source, and the rest $m = 12 - 2n$ classes are marked as shared classes. For each source domain, it contains private classes and shared classes. Two different source domains contain two different private classes.

For a three-source-complementary data set, under the category order of multiduplication data set, the first m classes are selected as shared classes. The rest $12 - m$ classes are divided into three parts, and each part is chosen as the private classes for each source domain. The number of classes m contained in shared classes ranges from 0 to 12. $m = 12$ means these sources sharing all the common 12 classes, and $m = 0$ means no shared class across these sources. The target domain remains unchanged, which contains all the 12 categories.



Fig. 2. Sample images of Merced, AID, NWPU, and PatternNet. For some categories, the images in NWPU are more diverse than that in other data sets.

For convenience, arrow "$\rightarrow$" is used to denote source domains transferring information to the target domain. Merced, AID, NWPU, and PatternNet are simplified as M, A, N, and P, respectively. For example, the two-source case transfer task from source domains NWPU and AID to target domain Merced can be denoted as $\{N, A\} \rightarrow M$. To distinguish the transfer of multiduplication data set and multicomplementary data set, $\bar{M}$, $\bar{A}$, $\bar{N}$, and $\bar{P}$ are used to denote the incomplete source domains in multicomplementary data set, respectively. Table II provides an example of two-source-complementary data set $\{\bar{N}, \bar{A}\} \rightarrow M$ with shared classes is set to 8. Table III provides an example of three-source-complementary data set $\{\bar{N}, \bar{A}, \bar{M}\} \rightarrow P$ with shared classes is set to 6.

B. Experimental Settings

1) Training Details: The model is implemented in a Pytorch environment. All the experiments are implemented in a workstation with GeForce GTX Titan X GPU, Inter Core i7-5930 K 3.50-GHz CPU, and 64-G RAM. In the experiment, all the images are resized to 224 $\times$ 224 pixels. Three RGB-bands are used as input. The model uses AlexNet [31] as backbone and is fine-tuned from AlexNet pretrained on the ImageNet data set [48]. The feature representation module is composed of top five convolution blocks in AlexNet. The classifier complement module is composed of K identical subnetworks, each containing three fully connected layers same as AlexNet. The cross-domain alignment module is composed of K discriminators, each of which is a two-output classification network composed of two fully connected layers. K is the number of source domains. The convolutional layers of the model are finetuned with top three convolutional layers frozen, and all fully connected layers are trained through backpropagation.

The model is trained via Adam. Momentum is set to $(0.9, 0.99)$. The learning rate starts from 10^{-4} and dynamically reduces by multiplying a factor of 0.1 after several epochs. Batch size is carefully chosen by experiment on three-source-complementary data set. The results with different batch sizes on four different transfer tasks are shown in Fig. 3. It can be seen from Fig. 3 that a larger batch size can

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TABLE IV
PERFORMANCE COMPARISON WITH STATE-OF-THE-ARTS ON TWO-SOURCE-COMPLEMENTARY DATA SET (SHARED CLASSES ARE SET TO 8)

	Model	$\{\bar{N}, \bar{A}\} \rightarrow M$	$\{\bar{A}, \bar{M}\} \rightarrow N$	$\{\bar{N}, \bar{M}\} \rightarrow A$	Average
Source combine	RevGrad [36]	63.40%	67.67%	74.59%	68.55%
	ADDA [26]	65.30%	65.32%	74.60%	68.40%
	Siamese-GAN [14]	77.00%	76.91%	81.06%	78.32%
	MSCN-AlexNet	65.05%	64.96%	74.61%	68.21%
	MSCN-CA	79.00%	75.12%	82.79%	78.97%
Multi-source	MSCN-CC	66.00%	54.45%	70.36%	63.60%
	MSCN (proposed method)	**81.92%**	**77.33%**	**84.44%**	**81.23%**

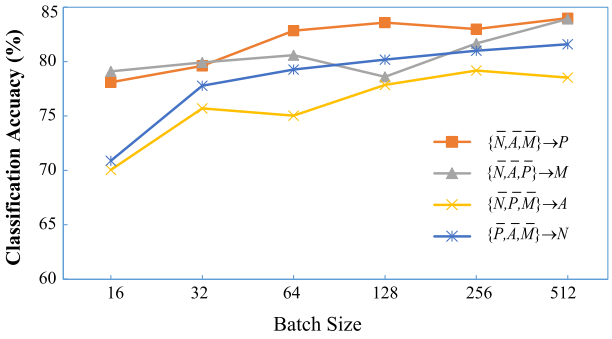


Fig. 3. Sensitivity analysis with respect to the batch size on three-source-complementary data set (shared classes are set to 6).

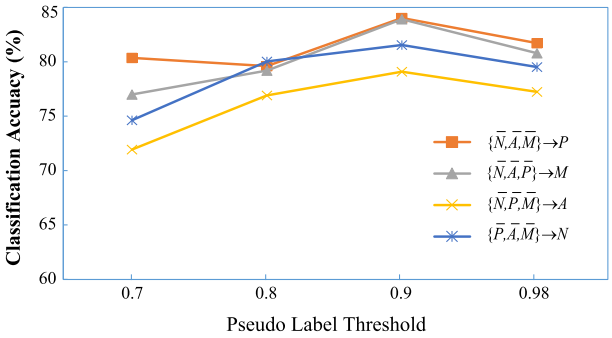


Fig. 4. Sensitivity analysis with respect to pseudolabel threshold on three-source-complementary data set (shared classes are set to 6).

achieve higher accuracy, but the accuracy will not continue growing too much when the batch size is larger than 256. Finally, batch size is set to 512 for balance. Pseudolabel threshold is also carefully selected. Classification results with different thresholds are shown in Fig. 4. The highest accuracies are obtained with threshold fixed at 0.9. Hence, pseudolabel threshold is set to 0.9 for all the subsequent experiments.

To assess the suitability of the proposed method, three variants of MSCN are designed as the baselines to illustrate the effectiveness of cross-domain alignment and classifier complement modules.

a) MSCN-AlexNet: This model is the variant that only contains AlexNet backbone. Cross-domain alignment and classifier complement modules are not contained in this variant. All images from multiple-source domains are combined into a single-source domain to train this model.

b) MSCN-CA: This model is the variant containing a cross-domain alignment module. The classifier complement module is not contained. Multiple-source domains are combined into a single-source domain as the model input.

c) MSCN-CC: This model is the variant containing classifier complement module. Cross-domain alignment module is not contained. This variant receives multiple-source domains as input and assigns each source domain a source-specific classifier.

MSCN-AlexNet and MSCN-CA both combine multiple sources to one source. The two variants are considered for two reasons. The first reason is that current researches only perform in the setting that any single-source domain covers all the categories of the target domain and have not considered the multisource compensation case. To make a fair comparison, multiple sources are combined into one source. The second reason is to verify the need for considering the distribution discrepancy across multiple sources. If multiple sources are combined to a single source, will the performance of single-source domain adaptation method outperform the multisource domain adaptation method?

2) Effectiveness of Cross-Domain Alignment: In order to evaluate the effectiveness of cross-domain alignment model, comparison between MSCN and the three variants is conducted on three transfer tasks $\{\bar{N}, \bar{A}\} \rightarrow M$, $\{\bar{A}, \bar{M}\} \rightarrow N$, and $\{\bar{N}, \bar{M}\} \rightarrow A$. Since different ratios between shared classes and private classes have a large impact on experimental accuracy, in this part, the shared classes are set to 8 for these experiments. A comparison of a different number of shared classes is analyzed later. The classification accuracies of shared classes set to 8 are displayed in Table IV.

From Table IV, it can be seen that MSCN-CA model has a large accuracy improvement compared with MSCN-AlexNet model, an improvement of 13.95%, 10.16%, and 8.18%, respectively, on three transfer tasks, so does MSCN compared with MSCN-CC model, 15.92%, 22.88%, and 14.08%, respectively. This is because the source domains and target domain are acquired over different regions with different sensors and have a large distribution divergence. The cross-domain alignment module can reduce the domain shift very well, so the model with cross-domain alignment module can achieve good performance on target domain.

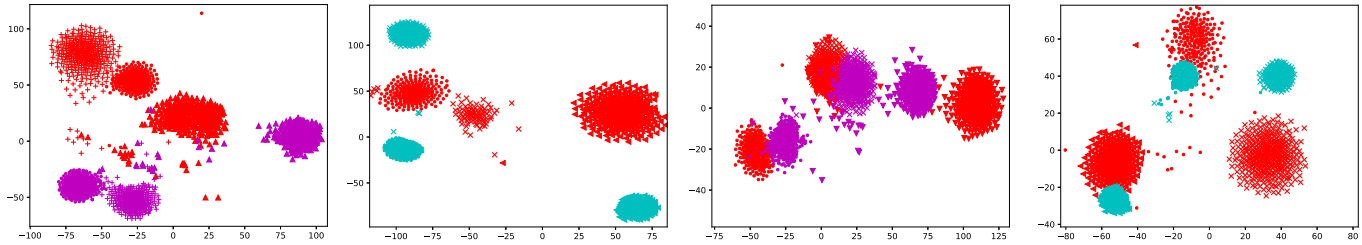
A feature distribution visualization on different domains is also conducted with the t-SNE [49]. As shown in Fig. 5, for

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TABLE V

PERFORMANCE COMPARISON WITH STATE-OF-THE-ARTS ON TWO-SOURCE-DUPLICATION DATA SET (A SPECIAL CASE OF TWO-SOURCE-COMPLEMENTARY DATA SET WITH SHARED CLASSES SET TO 12)

	Model	$\{N, A\} \rightarrow M$	$\{A, M\} \rightarrow N$	$\{N, M\} \rightarrow A$	Average
Source combine	RevGrad [36]	64.06%	68.59%	78.79%	70.48%
	ADDA [26]	67.52%	73.03%	81.31%	73.95%
	Siamese-GAN [14]	78.33%	79.36%	87.28%	81.66%
	MSCN-AlexNet	66.37%	73.14%	81.29%	73.60%
	MSCN-CA	79.42%	78.57%	90.19%	82.73%
Multi-source	MSCN-CC	73.92%	71.53%	82.23%	75.89%
	MB-Net [17]	72.25%	75.25%	77.58%	75.03%
	MSCN (proposed method)	**84.01%**	**79.55%**	**91.59%**	**85.05%**



(a) $\bar{N} \rightarrow A$, no domain adaptation. (b) $\bar{M} \rightarrow A$, no domain adaptation. (c) $\bar{N} \rightarrow A$, after domain adaptation. (d) $\bar{M} \rightarrow A$, after domain adaptation.

Fig. 5. Effectiveness of cross-domain alignment module on the distribution of different domain features. A visualization of features in $\{\bar{N}, \bar{M}\} \rightarrow A$ is shown with t-SNE [49]. The red color denotes target domain AID, magenta denotes source domain NWPU, and cyan denotes source domain Merced. (a) and (b) Features without domain adaptation. (c) and (d) Features after domain adaptation. By comparing (a) and (c), and (b) and (d), it can be found that source and target domains have a closer distribution after the cross-domain alignment.

the $\{\bar{N}, \bar{M}\} \rightarrow A$ task, the features obtained by MSCN-CC model have a large shift between source and target domains, but MSCN matches the distribution well with the help of cross-domain alignment module.

3) Effectiveness of Classifier Complement: From Table IV, it can also be noticed that assigning each source a source-specific classifier is of great value by comparing the MSCN and MSCN-CA model, an improvement of 2.92%, 2.21%, and 1.65%, respectively. MSCN considers the distribution divergence across multiple sources. MSCN-CA model that simply merges all sources together results in a negative transfer.

Please note that the reason for the MSCN-CC model performing worse than MSCN-AlexNet model lies in the classifier. The MSCN-AlexNet model combines all the source domains in one domain and trains a 12-categories-classifier. However, the MSCN-CC model is composed of two ten-category classifiers. The private classes of one source domain are falsely classified into ten irrelevant categories by another source-specific classifier. Luckily, the target images are classified with multiple classifiers, so other source-specific classifiers can make up for the falsely classified categories. Finally, the accuracies of MSCN-CC model are slightly lower than MSCN-AlexNet model but also acceptable.

This situation also exists in the MSCN model, but the impact is weak. With large distribution divergence existing across different domains, source-specific classifier in the MSCN-CC model has low confidence to assign each image to the corresponding category. The target classifier can be easily disturbed by the wrong classification predictions. However, in MSCN model, with the cross-domain alignment module reducing the domain shift, the source-specific classifier has high confidence in the corresponding classes. The wrong classified disturbance has less impact on the final classification decision. Moreover, when each source domain shares same classes with target domain, the classification accuracies also verify the necessity of assigning different sources a specific classifier, as analyzed later.

C. Experiment on Two Source Case Cross-Domain Data Set

1) Comparison on Two-Source-Duplication Data Set: The conventional multisource domain adaptation is a special case of the MSCN method when every source domain has the same categories with target domain. A comparison is made between our method and state-of-the-art cross-domain scene classification methods on two-source-duplication data set. One multisource domain adaptation method, MB-Net [17], and three single-source adaptation methods, (RevGrad) [36], adversarial discriminative domain adaptation (ADDA) [26], and Siamese-GAN [14], are compared. MB-Net aims to learn transfer knowledge from multiple-source domains by mitigating the data shifts at middle feature level and decision level. RevGrad reduces the domain shift by making the source and target domains indistinguishable for a domain classifier. ADDA learns the domain-invariant features by combining

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TABLE VI
PERFORMANCE COMPARISON WITH STATE-OF-THE-ARTS ON THREE-SOURCE-COMPLEMENTARY DATA SET (SHARED-CLASSES IS SET TO 6)

	Models	$\{\bar{N},\bar{A},\bar{M}\}\rightarrow P$	$\{\bar{N},\bar{A},\bar{P}\}\rightarrow M$	$\{\bar{N},\bar{P},\bar{M}\}\rightarrow A$	$\{\bar{N},\bar{A},\bar{P}\}\rightarrow M$	Average	Training Time(s)	Test Time(s)
Source combine	RevGrad [36]	68.75%	66.37%	62.83%	67.35%	66.33%	257	1.61
	ADDA [26]	68.99%	60.39%	62.39%	65.08%	64.21%	188	2.20
	Siamese-GAN [14]	**84.50%**	81.42%	77.17%	80.58%	80.92%	181	11.21
	MSCN-AlexNet	72.15%	70.94%	71.71%	71.55%	71.59%	109	1.52
	MSCN-CA	80.83%	72.33%	73.58%	75.75%	75.62%	119	1.57
Multi-source	MSCN-CC	77.33%	74.91%	71.50%	67.17%	72.73%	232	1.70
	MSCN (proposed method)	83.91%	**83.83%**	**79.08%**	**81.50%**	**81.97%**	240	1.68

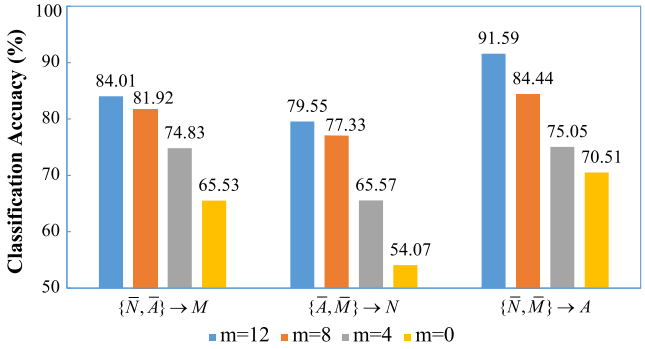


Fig. 6. Classification accuracy (%) of MSCN with different shared classes m on two-source-complementary data set.

discriminative modeling, untied weight sharing and GAN loss. Siamese-GAN learns the domain-invariant features by the adversary between an encoder–decoder architecture and a discriminator. MB-Net is a multisource domain adaptation method; it receives multiple sources as input. For the single-source adaptation methods, all the source domains are merged into a single source as the model's input. All comparison methods are carefully conducted with the released code by the corresponding authors or the settings defined in the original articles. For RevGrad and ADDA, since the data sets used in original articles have a large difference with the data sets used in our experiments, we changed the learning rate and batch size by referring to related article [14], [17], [26], [36]. For Siamese-Gan and MB-Net, the parameters are the same with settings in the original articles [14], [17].

The results are presented in Table V. From Table V, it can be seen that MSCN achieves the best accuracies on three tasks, 84.01%, 79.55%, and 91.59%, respectively. The results show the effectiveness of the cross-domain alignment module and the importance of assigning a source-specific classifier to each source domain in multisource classification task.

2) Comparison on Two-Source-Complementary Data Set: In this part, a comparison between MSCN and some state-of-the-art methods is conducted. Furthermore, the influence of a different number of shared categories on classification accuracies is analyzed. The experiments are conducted on two-source-complementary data set.

a) Comparison to state-of-the-art methods: Since multi-source complementary setup differs from conventional setup, most of the existing domain adaptation methods cannot directly apply to this problem. In this part, single-source domain adaptation methods are used as comparative methods by combining multiple sources into a single source. Three state-of-the-art methods, RevGrad [36], ADDA [26], and Siamese-GAN [14], are selected. The classification accuracies comparison is displayed in Table IV. It can be seen from Table IV that the MSCN method performs well in three transfer tasks and achieves better results than three state-of-the-art methods.

It should be noticed that most methods have the highest classification accuracies on transfer task $\{\bar{N},\bar{M}\}\rightarrow A$ and lowest classification accuracies on transfer task $\{\bar{A},\bar{M}\}\rightarrow N$. The reasons for this are in two parts. First, the similarity between source domain and target domain has large impact on classification accuracy. It is hard to transfer sufficient information to target domain if a source domain has large divergence with target domain. For example, for the task $\{\bar{A},\bar{M}\}\rightarrow N$, category Airfield is only contained in source domain AID, so the transfer of category Airfield is $\bar{A}\rightarrow N$. However, the images of category Airfield in NWPU are more diverse than the ones in AID, as shown in Fig. 2. As a result, the information transferred from AID is insufficient, which leads to a low classification accuracy on category Airfield. There is also the case for category Farm and some other categories. For the task $\{\bar{N},\bar{M}\}\rightarrow A$, transfer of category Airfield is $\bar{N}\rightarrow A$. The information of category Airfield contains in NWPU is sufficient, so the classification accuracy is high. Second, the amount of target domain images also has an influence on classification results. The NWPU data set has 10 500 images, the Merced and AID have small amounts of 1200 and 4710, respectively. A large amount of images leads to low classification accuracy on NWPU. To control the effect of different numbers of images, every class is set to contain 100 images in the three-source case cross-domain data set. A detailed analysis is provided later.

b) Analysis on a different number of shared-classes: The performance of MSCN has a large relation with the number of shared-classes among source and target domains. Fig. 6 shows the results with different shared-classes on three tasks.

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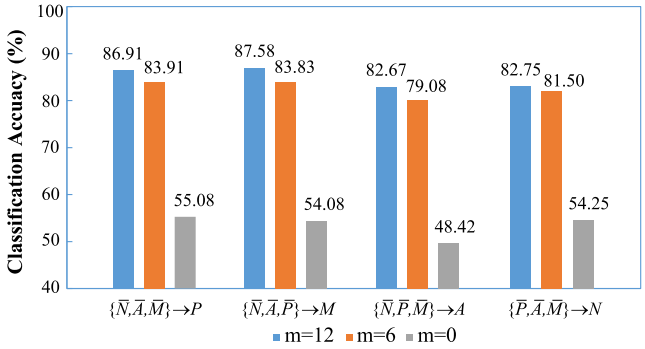


Fig. 7. Classification accuracy (%) of MSCN with different shared classes m on three-source-complementary data set.

The more shared classes among source and target domains, the higher the classification accuracies on the target domain. When shared classes are set to 12, the target data set has the highest accuracies of 84.01%, 79.55%, and 91.59% on $\{\bar{N}, \bar{A}\} \rightarrow M$, $\{\bar{A}, \bar{M}\} \rightarrow N$, and $\{\bar{N}, \bar{M}\} \rightarrow A$, respectively. When the shared classes are set to 0, the target domain has the worst classification accuracies.

The reason for this is obvious. When the categories of each source domain decrease, the categories source-specific classifier can classify also decreases. More target domain images are misclassified to irrelevant categories, which resulting in a low classification accuracy.

D. Experiment on Three-Source Case Cross-Domain Data Set

To prove the generality of the proposed method, experiment on three-source case cross-domain data set is conducted. First, classification accuracy is compared with state-of-the-art methods on three-source-complementary data set. Second, the training time and test time for these methods are provided. Finally, for the proposed method, classification accuracy with different numbers of shared classes is analyzed.

As shown in Table VI, our method achieves the best performances on most transfer tasks. By comparing the four transfer tasks, a conclusion similiar to the two-source case can be obtained. A task whose source and target domains have large similarity will achieve high classification accuracy.

The training and test time is also provided in Table VI. Time is counted in seconds. The proposed method and other compared methods are conducted under the same condition. Our method can converge in an acceptable time during training. Test time is also comparable to other methods. Siamese-GAN has a relatively long test time because it extracts features using VGG16 network [50]. Other methods use Alexnet as the backbone. The training time of Siamese-GAN is comparable because the parameters of VGG16 are not optimized during training.

The performance of our method with different shared classes is given in Fig. 7. When the shared classes are set to 12 (i.e., the three-source-duplication data set), the best performance is achieved. Same as the observation on the two-source case, performance decreases as the number of shared categories decreases. Moreover, since the number of images per class is set to 100 in these experiments, the influence of different numbers of images is eliminated.

V. Conclusion

In this article, we focus on remote sensing cross-domain scene classification using multiple complementary sources. An MSCN is proposed to solve the domain shift and category mismatch problems. First, a pretrained CNN is exploited to learn the feature representation for each domain. Second, a cross-domain alignment module is developed to reduce the distribution discrepancy between each source and target domains. Finally, a classifier complement module is designed to align the categories in multiple-source domains and learn a target domain classifier. Extensive experiments are conducted on four heterogeneous remote sensing scene classification data sets to validate the effectiveness of the proposed method. In future work, we plan to tackle a more challenging scenario: classifying the target category that is not present in any source domains by other effective strategy.

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